XE

XELOR · TECHNICAL STRATEGY

Technology Stack & Production Roadmap

What the demo actually runs, what production still needs, the extension points already reserved, and a pragmatic target stack for a dependable manufacturing product.

Prepared from the repository, module/API audit, deployment package, managed-service reference deck and primary standards

8 August 2026

1. Executive decision

XELOR has a strong product-engineering foundation for a demo: a TypeScript monorepo, a boundary-enforced modular monolith, PostgreSQL with forced tenant isolation, Keycloak identity, durable approval/audit patterns, event delivery, a real Next.js application, and a governed AI spine. The current build also includes a cross-industry Decision & Risk Commander, durable evidence links and a verified-outcome ledger. The right production move is to harden and operate this architecture—not rewrite it or split it into microservices.

> **Recommended direction:** retain the modular monolith through product-market fit; deploy separate Web, API and Worker containers; move state to managed AWS services in Mumbai; add observability, recovery, security automation and CI/CD; close the manufacturing workflow gaps before adding architectural complexity.

> **Security and reliability reality:** no honest architecture is unbreachable or incapable of failure. The production target is a resilient, recoverable, least-privilege system with defence in depth, small blast radii, tested restoration, continuous detection and measurable service objectives.

What is unusually good

- Tenant context is derived from verified identity and reinforced by PostgreSQL FORCE RLS.
- Ledger-critical writes stay transactional and use controlled write paths.
- Audit, approvals and AI actions are append-only and tamper-evident.
- Domain boundaries and provider ports leave credible exit paths.
- The demo is deterministic and rebuildable instead of relying on fragile fake screens.

What blocks production

- A Railway demo package exists, but no production cloud environment, autoscaling or managed recovery is operating.
- Telemetry, alerting, incident response and SLOs are not yet operational.
- Several core manufacturing workflows remain incomplete.
- Backup claims need automated restore drills and evidence.
- Security assurance, load testing, accessibility and release governance need formal gates.

Status language used in this report

LIVE wired and exercised in the prototype **PARTIAL** real foundation but incomplete production behavior **RESERVED** a deliberate interface/schema extension point **RECOMMENDED** proposed after the demo.

5	31	242	9/9
workspace packages/apps	NestJS controller surfaces	migration CREATE TABLE declarations	Agent OS identities connected

Counts are repository observations, not a claim that every table or endpoint is complete for commercial production.

2. Stack used to build and run the demo

Layer	Technology in repository	Status	What it does today
Runtime & language	Node.js 22; TypeScript 5.7; ESM	LIVE	One language across browser, API, platform rules, tests and workers.
Monorepo	pnpm 9 workspaces	LIVE	Coordinates five workspace packages: `apps/web`, `apps/api`, `apps/edge`, `packages/platform` and `packages/db`.
Web application	Next.js 15 App Router; React 19; Tailwind CSS 4; Radix UI; Lucide; Three.js	LIVE	Authenticated XELOR shell, horizontal module tabs, global Human Approvals entry with live pending count, dedicated approval inbox, department gateway, ONYX rail, Mission Control and an adaptive 3-D digital-twin sign-in scene.
API	NestJS 11; Express adapter; RxJS; SWC build	LIVE	Boundary-oriented modular monolith with 31 controller surfaces, including health/readiness.
Validation	Zod 3	LIVE	Validates most request bodies and agent capability inputs; a few documented endpoints still bypass the pattern.
Database	PostgreSQL 17; pgvector; pg_trgm; btree_gist	LIVE	Transactional ERP system of record, vector-ready columns, fuzzy master matching and constraint-backed invariants.
Data access	Drizzle ORM 0.36; SQL migrations; node-postgres	LIVE	Typed schema access plus explicit migrations and operational checks.
Tenancy	AsyncLocalStorage context + `SET LOCAL` + PostgreSQL FORCE RLS	LIVE	Fail-closed shared-schema tenant isolation; portal adds customer-account scoping.
Identity	Keycloak 26; OIDC Authorization Code + PKCE; jose/JWKS; in-app RBAC; explicit public-demo mode	LIVE	Normal mode derives signed tenant claims and permissions from Keycloak. A separately configured, sign-in-free hosted-demo mode maps only fixed seeded personas and is restricted to disposable investor data.
Queue & cache	Valkey 8; BullMQ 5; ioredis	LIVE	Outbox relay and idempotent asynchronous consumption; local cache/broker foundation.
Audit & workflow	Hash chains; append-only tables; W1 approval engine; Agent OS human gates	LIVE	Tenant-scoped pending inbox, permission-checked Approve/Reject, mandatory decision note, identity/time attribution, durable pause/resume and tamper evidence.
AI	Thin provider router; deterministic stub; optional Ollama; custom eval harness; Agent OS; Decision Commander	PARTIAL	Cross-module analysis, missions and approvals are live; language reasoning remains deterministic and no hosted provider is active.
Decision data	PostgreSQL evidence-link graph and outcome metrics with RLS, audit and explicit verification state	LIVE	Estimated exposure and verified value are stored and reported separately; verification requires observed data, method and named reviewer.
Testing	Node test runner; Playwright 1.55; ESLint 9; TypeScript checks	LIVE	783 unit/service tests pass (712 platform, 4 database, 56 API, 6 edge and 5 web-client); the live investor matrix passes 99/99 and Playwright lists 32 browser scenarios separately.
Deployment packaging	Docker Compose plus Railway Dockerfiles, service JSON, health/readiness and boot scripts	LIVE	Local seven-service development and a five-service hosted-demo topology are reproducible. A currently operating cloud environment or production topology is not claimed.

Branding note: internal package, database and Keycloak realm identifiers still use the historical deployment slug in places. They are technical contracts, not rendered product copy. Renaming them is possible but should be a separately rehearsed infrastructure migration—not a search-and-replace during demo hardening.

3. Architecture that exists now

Next.js / React Web shell → NestJS modular API → PostgreSQL 17 FORCE RLS

Keycloak OIDC + claims → Tenant + RBAC guards → Domain services single write paths

Transactional outbox → Valkey + BullMQ relay/worker → Idempotent consumer effect

Human-in-the-loop architecture now visible in the product

Agent reaches consequence gate → Durable pending approval row → Global counter + approval inbox

Human reviews exact consequence → Written note + Approve / Reject → Audit event + resume or cancel

- **Discoverable everywhere:** authorised users see a labelled Approvals control in the global top bar; pending work carries a count and attention state.
- **One real source:** the counter and inbox read the tenant-scoped Agent OS approval table, not a second presentation-only queue.
- **Deliberate decision:** approval requires a written note; the user identity, note, decision and time are retained in the operational and audit records.
- **Narrow authority:** approval releases only the exact proposed action. It does not grant wider agent access or approve future missions.
- **Fail-safe behavior:** rejection cancels the mission; approval resumes from the persisted graph state and side-effect ancestry is verified again.

New domain operating surfaces

Module	Agent owner	Implemented user surface	Current data truth
Working Capital	RASP Finance & Working Capital	Overview, money in/out, stock holding cash, margins, 13-week forecast, scenarios and finance readiness pack.	Agent capabilities read tenant-scoped Accounts data; module KPIs and action stories remain curated demo scenarios until forecast/collection persistence is added.
QMS & Audit	KILN Operations & Quality	Overview, live inspections, NCR, containment, root cause, CAPA, controlled documents, audits, training and evidence packs.	Inspections and the Northstar NCR/CAPA thread are persisted system-of-record reads; document/audit breadth remains explicitly illustrative.
Human Approvals	HEXA governance + named human	Global counter, dedicated inbox, consequence preview, mandatory note, Approve/Reject and Mission Control handoff.	Live Agent OS pending records and decision API; permission and tenant checks are enforced server-side.

Why the modular monolith is the correct near-term shape

- Manufacturing transactions cross inventory, quality, production, purchasing and accounts; keeping correctness in one transactional boundary is currently more valuable than independent service scaling.
- Module boundaries are already explicit. Cross-module access uses ports/services or outbox events, giving a future extraction path without paying distributed-systems cost now.
- API, web and worker processes can scale independently even while sharing one codebase and database.
- The current team can debug one deployment and one data model while the product and workflows are still moving quickly.

Production topology recommended after the demo

CloudFront + WAF **public edge**	→	**ALB** **TLS routing**	→	**ECS Fargate** **Web / API / Worker**
Keycloak HA **or managed IdP**	→	**RDS PostgreSQL** **Multi-AZ**	→	**S3 + KMS** **documents/backups**
Valkey managed **queue/cache**	→	**OTel Collector** **traces/metrics**	→	**CloudWatch/Grafana** **alerting**

> **Do not introduce Kubernetes yet.** ECS/Fargate fits three container workloads without a control-plane burden. Reconsider EKS only when team size, workload diversity, GPU scheduling, multi-region policy or platform engineering requirements justify it.

3A. Module maturity map

All 22 modules are registered and visible, but “module exists” is not the same as “enterprise workflow complete”. This map names the real centre of each module and the missing edge that matters most.

Owner	Module	Working centre	Most important boundary
ONYX	Copilot LIVE	21 permission-checked read intents, fixed queries, evidence, refusals and history.	No unrestricted Q&A or write action; hosted narration is off.
ONYX	Agent OS LIVE	7 graphs, 19 capabilities, parallel waves, checkpoints, approvals and recovery.	Language reasoning is deterministic; dispatch creates a work item, not ERP completion.
ONYX	AI Operations PARTIAL	Registry, provider policy, kill/budget, cost, evaluations and action logging.	No live hosted provider; implementation/golden-set coverage is incomplete.
HEXA	Organisation LIVE	Companies, registrations, departments and transaction-safe numbering.	No consolidation or corporate-secretarial lifecycle.
HEXA	Administration PARTIAL	Roles, typed permissions, SoD, incident/privacy clocks and audit verification.	Inactive roles/grants fail closed; row/field policy remains mainly preview and escalation/delegation needs depth.
RELAY	Managed Services PARTIAL	Four-stage lifecycle, responsibility map, service catalogue, incident/change/review workspace and governed assurance graph.	Illustrative data only; live ITSM, telemetry, service tables, staffing and contractual SLOs remain production work.
HEXA	Integration PARTIAL	Retry/DLQ/circuit/signature/statutory state plus Factory Connect gateways, robot/AMR bindings, submitted state/location/dwell records and governed command evidence.	The command endpoint performs simulator policy evaluation only: it neither calls `apps/edge` nor records a controller acknowledgement. OPC UA, MQTT, ROS 2, Cisco Spaces and Splunk are catalogue/reserved targets; certified adapters, gateway identity and physical transport remain deployment work.
MICA	Sales PARTIAL	Customer, GST order, credit gate, dispatch, invoice and receivable.	No lead/opportunity/quotation or complete cancellation/reservation workflow.
MICA	Customer Care & Warranty PARTIAL	Manufactured-product cases, account isolation, response targets, entitlement, complaints, spares and customer feedback.	No integrated outbound transport; warranty rows are not created by normal flow. XELOR application, integration and AI service incidents are owned separately by RELAY.
SPAR	Purchase PARTIAL	Vendor, two-step approval, PO and atomic goods receipt.	Cycle ends at GRN; no supplier invoice, match, AP or payment.
SPAR	Inventory PARTIAL	Single append-only stock path, locks, FIFO batches and on-hand.	No bin/reservation/count/serial genealogy or complete valuation.
AXLE	Engineering PARTIAL	Item and BOM foundations consumed by Sales, Planning and Production.	No ECR/ECO change intent, impact, approval or effectivity.
AXLE	Planning PARTIAL	MRP, calendars, lot sizing, pegging, immutable workings and exceptions.	Capacity is infinite and scheduling is a draft heuristic.
KILN	Production PARTIAL	Pinned components, predecessor-gated operations, issue/receipt plus additive robot-cell and AMR evidence views.	No physical controller is connected; safety PLCs, interlocks and emergency stops remain locally authoritative.
KILN	QMS & Audit PARTIAL	Inspections, snapshotted limits, disposition, NCR/CAPA and evidence.	Auto-created inspections, calibration and broader document/audit lifecycle.
KILN	Maintenance PARTIAL	Assets, requests, work, downtime, PM and reliability calculations.	No sensors, complete shift calendars or predictive engine.
RASP	Accounts PARTIAL	Balanced append-only GL, reversal, trial balance and receivables.	No period close, AP, bank reconciliation or full statements.
RASP	Employee Spend PARTIAL	Budget reservations, claims/receipts, approvals and tax gates.	Prepared posting is not fully wired to Accounts; cost-centre masters differ.

Owner	Module	Working centre	Most important boundary
RASP	People **PARTIAL**	Employees, punches, attendance and dated statutory payroll.	No broad recruitment/performance/learning or device integration.
RASP	Working Capital **PARTIAL**	Real cash-position read and a governed cross-agent review mission.	Most screen values and three finance tools are illustrative wrappers.

3B. Transaction ownership and cross-module dependencies

The architecture works because cross-module flows do not bypass record ownership. The following boundaries are the highest-value contracts to preserve during future implementation.

Business event	Owning transaction	Cross-module effect	Failure behavior
Customer dispatch	Sales coordinates through the Stock and Accounts ports.	Inventory goods-out, delivery note, invoice and receivable commit together.	Credit or stock refusal aborts all of it; a retry uses the same idempotency key.
Supplier receipt	Purchase validates approved outstanding PO quantity.	GRN and Inventory receipt share one transaction.	Over-receipt or stock posting fault leaves neither a GRN nor a movement.
MRP run	Planning gathers Sales/Engineering/Inventory/Purchase/Production inputs through ports.	Stores workings, pegs, planned orders and exceptions as one dated result.	Cycles/calendar faults refuse the run; completed workings cannot be edited.
Production release/completion	Production pins the Engineering BOM and operation sequence.	Inventory issues components and receives finished goods; Quality evidence is linked separately.	Open predecessors, insufficient stock or completion blockers return without pretending progress.
Quality disposition	Quality owns the verdict and applied spec/sampling snapshots.	Inventory owns quarantine/release movement; finding and CAPA preserve follow-up.	An executed disposition without its movement reference is structurally refused.
Employee claim	Employee Spend locks and moves the budget reservation ledger.	People supplies identity/grade; AI Operations may govern extraction; Accounts receives a prepared posting.	Unbudgeted/override/tax ambiguity stays visible; current Accounts hand-off is incomplete.
Payroll approval	People calculates from attendance and dated statutory tables.	Accounts posts the balanced payroll voucher synchronously.	Missing rate or duty conflict stops the run; payroll and journal cannot disagree.
Agent-controlled action	Agent OS owns the graph, approval and governed work item.	The specialist receives an assignment referencing the exact approved payload.	No approved ancestor means no dispatch; dispatch never masquerades as domain completion.

Extraction rule for future microservices: do not split a module merely because it has many endpoints. Extract only when there is a measured scaling, ownership, deployment or fault-isolation need—and preserve each atomic contract with an explicit saga/reconciliation design.

3C. Railway demo deployment now present

The repository includes deployment packaging for a public demonstration, not a claim of production operations. Railway is a better fit than Vercel alone because API, worker, PostgreSQL and Valkey must accompany the Next.js front end.

Public visitor **Railway URL**	→	**xelor-web** **only public service**	→	**xelor-api** **private service**
xelor-worker **outbox consumer**	→	**xelor-postgres** **durable volume**	→	**xelor-valkey** **demo coordination**

Concern	Implemented demo behavior	Still required for production
Images/config	Four Dockerfiles, five Railway service descriptors, watch patterns and health/restart policies under `infra/railway`.	Registry signing, SBOM, vulnerability policy, promoted immutable digests and IaC.
Database boot	Wait for PostgreSQL, apply 90 forward migrations through `0091`, start API, seed/checkpoint a named demo version once, then report readiness.	Managed Multi-AZ database, migration rehearsal/rollback policy, PITR and restore evidence.
Identity	Explicit Web/API public-demo flags provide fixed seeded personas and remove the sign-in page for the disposable demo.	Keycloak/enterprise IdP, MFA, lifecycle, strict session policy and zero demo bypass.
Networking	Only web is public; API, database, worker and Valkey communicate on Railway private DNS.	WAF/edge protection, network policy, private endpoints, rate limits and threat monitoring.
Operations	Liveness/readiness, dependency waits, restart policy and reproducible local five-container parity Compose.	Metrics/traces/log aggregation, SLOs, alerting, on-call, capacity and incident response.
Cost	Fits Railway's five-service trial limit and keeps AI on the deterministic stub.	Funded always-on environment; five services are unlikely to remain inside the long-term free allowance.

> **Current state:** deployment-ready demo packaging exists in the working directory. No generated Railway URL, active project, TLS certificate or DNS record can be proven from repository files alone.

3D. RELAY managed-service stack and implementation path

The new module deliberately begins as one shared operating-model definition, one read API, five presentation screens and one bounded graph. This proves ownership and workflow before the team adds a second operational database or promises service coverage.

Customer outcome catalogue + entitlement → **Signals + contact OTel / ITSM adapters** → **RELAY incident + change** → **Specialist technical action** → **Review SLO + improve**

Layer	MVP now	Production implementation	Control
Service model	Typed shared lifecycle/responsibility/demo snapshot.	Tenant-scoped service, entitlement, SLO, incident, event, request, problem, change, communication, review and improvement tables.	RLS, immutable event history, retention and one accountable owner.
Observability	Explicitly seeded examples.	OpenTelemetry Collector and provider-neutral trace/metric/log ingestion; component-to-service mapping.	Redaction, sampling policy, provenance, freshness and cost limits.
Service management	XELOR-only five-screen workspace.	ITSM, paging and customer-channel adapters implemented through HEXA Integration with external IDs and idempotency.	RELAY coordinates; HEXA owns connectors and security; specialists own fixes.
Automation	Read capability plus approval-bound coordination work item.	Start read/recommend/draft; later add reversible low-risk capabilities with postconditions.	Permission intersection, approval ancestry, retries, audit and kill switch.
People	Role model documented.	Named service owner, manager, transition lead, desk/operations lead, technical owners, on-call and customer approvers.	Never sell 24×7 until shifts, backups and escalation drills are evidenced.

Recommended rollout

1. One pilot customer, business-hours coverage and 3–5 customer outcomes.
2. Agree catalogue, responsibility matrix, severity and calculation rules before coding integration.
3. Add tenant data model and RLS, then ITSM/OTel adapters in shadow mode.
4. Rehearse incident, change, restore and exit; measure SLOs without contractual credits.
5. Expand coverage and automation only after staffing and evidence support the promise.

> **Current label must remain:** illustrative managed-service operating model. It is not evidence of live telemetry, a staffed desk, external communications or contractual availability.

4. Stack needed immediately after the demo

Capability	Recommended stack	Why it is required	Exit criterion
Container delivery	Docker multi-stage images, Amazon ECR, ECS/Fargate, ALB, CloudFront, WAF	Immutable releases, health checks, horizontal scaling and safe external exposure.	Rolling or blue/green release with tested rollback and no manual server mutation.
Managed database	RDS PostgreSQL Multi-AZ, encrypted storage, PITR, parameter groups; RDS Proxy or PgBouncer only after measurement	Failover, backups and operational ownership suitable for a system of record.	Documented RPO/RTO met in a restore/failover exercise.
Managed broker	Amazon ElastiCache for Valkey or an equivalent managed Valkey service	Removes single-host queue/cache risk while preserving BullMQ semantics.	Queue replay, poison message and broker-failover tests pass.
Object/document storage	S3, KMS, presigned upload/download, antivirus scan queue, lifecycle policies	Receipts, certificates, drawings, exports and evidence must not live on container disks.	Every document has tenant ownership, checksum, retention and access audit.
Secrets	AWS Secrets Manager + KMS; short-lived workload identity; no static cloud keys	Database, OIDC and connector credentials need rotation and access evidence.	Production contains no secret in source, image, task definition or developer shell history.
Infrastructure as code	OpenTofu modules; remote encrypted state; environment/account separation	Reproducible Mumbai primary and Hyderabad recovery environments.	A clean account can be built from reviewed code.
Production CI/CD	Extend the live GitHub verification workflow with cloud OIDC federation, SBOM, signing, deployment, promotion and rollback	Turns the current test/container/demo gate into a complete release policy.	No production deploy bypasses the pipeline; provenance is retained.
Observability	OpenTelemetry Node instrumentation and Collector; CloudWatch/Grafana; Sentry or equivalent error tracking	Correlates browser, API, database, queue and AI behavior without reading raw production data.	SLO dashboards and actionable alerts cover the critical workflows.
Security edge	WAF, Shield baseline, private subnets, VPC endpoints, GuardDuty, Security Hub, CloudTrail	Reduces attack surface and supplies security evidence.	Threat model, penetration test and critical finding remediation completed.
Notifications	SES/SNS initially; provider abstraction for WhatsApp/SMS vendors	Approvals, escalations, customer/service events and operational alerts need reliable delivery.	Templated, tenant-aware, consent-aware delivery with bounce/retry evidence.

5. Space deliberately left for future stacks

Existing extension point	Current implementation	What can plug in later	Trigger to adopt
`AiProvider` / thin router	Deterministic stub or Ollama by configuration	OpenAI Responses API adapter, Amazon Bedrock Converse, Gemini or another approved provider	A feature passes its eval/grounding/privacy gate and has an agreed data-processing posture.
`WorkflowExecutor`	In-house W1 approvals	Temporal for long-running, cross-system workflows	Workflows last days, require many compensations/signals, or W1 operations become costly.
Outbox/event envelope	PostgreSQL → BullMQ/Valkey	Kafka/MSK, EventBridge or CDC for higher fan-out and independent consumers	Sustained throughput/fan-out or external analytics proves the current relay insufficient.
Domain ports	In-process Nest services	Extracted services behind HTTP/gRPC/events	A module needs independent ownership, scaling, release cadence or isolation—and the cost is measured.
pgvector columns/indexes	Extension enabled; knowledge-base vector field provisioned	Authorized RAG over manuals, drawings, SOPs and cases; external vector DB only if scale requires	Document ingestion, permissions and retrieval evals are ready.
Gotenberg	Local HTML-to-PDF container	Production document service with templates, signatures and archival	Quotes, invoices, payslips, certificates and compliance exports require governed rendering.
Keycloak OIDC	Local realm, personas and custom theme	SAML/LDAP federation, enterprise organizations, MFA/WebAuthn, SCIM provisioning	First enterprise customer identity requirements.
Integration/webhook module	Signed webhooks, retry/DLQ, statutory demo connectors	GSP, bank, logistics, payment and supplier adapters through connector workers	Contract, sandbox certification and reconciliation design are complete.
Analytics read path	Operational queries on PostgreSQL	Read replica, CDC to S3/Iceberg and a warehouse/query engine	BI queries materially affect ERP latency or cross-tenant analytics is required.

> **Architectural advantage:** the repository already contains more credible exit paths than the demo needs. The discipline is to adopt each new stack only when its operational trigger occurs.

6. Product gaps that matter more than new infrastructure

The repository's capability-gap register was produced by trying to execute the Northstar story. These are product workflow issues, not presentation polish.

Priority	Gap	Production impact	Recommended treatment
Resolved	Batch/lot consumption	FIFO allocation now consumes batch-tracked receipts through the single stock write path and records explicit splits.	Keep the allocation and insufficiency regression tests in the release gate; extend genealogy reporting for production.
Resolved	Idempotency/validation consistency	Database conflicts map to safe 409 responses; the identified maintenance and expenditure paths validate and deduplicate requests.	Retain conformance probes in the 99-check investor acceptance matrix.
P1	Quotation and opportunity	The commercial process starts at a committed sales order.	Add lead/opportunity, quote revisions, validity, approvals and quote-to-order conversion.
Resolved	Production operations	Northstar production reports four predecessor-gated operations with operator, time, quantities and evidence.	Deepen labour/machine booking and WIP analytics for production deployments.
Resolved	Persisted NCR/CAPA	The rejected inspection has a persisted NCR, containment, root cause and CAPA whose effectiveness review remains human-owned.	Extend toward full 8D, attachment lifecycle and customer-specific approval routes.
P1	Supplier invoice/AP/payment	Direct-material purchase cycle stops at goods receipt.	Supplier invoice, three-way match, exceptions, AP ageing and controlled payment run.
P1	Engineering change control	BOM version exists but change intent, impact and effectivity are not governed.	ECR/ECO, where-used impact, stock disposition, approvals and effective-from revisions.
Resolved	Employee Spend UI	Claims and budget decisions are operable in a registered, permission-checked web module.	Expand travel/advance/indirect-spend screens as pilot scope requires.
P2	Cost-centre master	Employee and budget vocabularies do not align.	Create one governed organizational/cost-centre master referenced by both modules.

Keep the boundary explicit. Quotation/opportunity, engineering change control and supplier AP remain production-roadmap gaps; do not claim end-to-end manufacturing ERP completeness yet.

7. Phased production roadmap

Phase 0 — Demo confidence (now to 4 weeks)

- Keep the authenticated route crawl, real Keycloak login, Decision Commander and end-to-end approval-inbox tests in the release suite.
- Measure pending-approval age, decision turnaround, recovery outcome and rejection reasons; add escalation without bypassing authority.
- Extend the evidence graph through governed connectors while retaining source ownership and tenant fencing.
- Produce signed container builds and keep the isolated top-level demo reset as a release check.
- Define data classification, tenant onboarding/offboarding and a minimum threat model.

Phase 1 — Pilot-ready foundation (1 to 3 months)

- Deploy Web/API/Worker on ECS/Fargate in Mumbai with RDS Multi-AZ, managed Valkey, S3/KMS and Secrets Manager.
- Implement OpenTofu, CI/CD, SBOM/signing, migrations-as-a-gate and environment promotion.
- Add OpenTelemetry, error tracking, database/queue dashboards and on-call runbooks.
- Enable MFA, session policy, rate limits, WAF, audit exports, restore drills and customer-isolated test environments.
- Close quotation/opportunity so a pilot follows the commercial process before commitment.

Phase 2 — Controlled production (3 to 9 months)

- Deepen operation execution and NCR/CAPA breadth; add AP/three-way match and engineering change control.
- Add document/object lifecycle, connector certification, reconciliation queues and customer notification channels.
- Run performance, chaos/failover, accessibility, mobile-responsiveness and penetration-test programs.
- Introduce a hosted AI provider only for features that pass offline and shadow-mode evaluation.

Phase 3 — Scale and ecosystem (9 to 18 months)

- Add read replicas/analytics CDC only after workload evidence; introduce Temporal only for workflows that exceed W1.
- Enterprise federation, SCIM, dedicated-tenant deployment option and multi-region recovery rehearsal.
- Public partner APIs, connector SDK/contracts, metering/billing and customer-facing status/audit exports.

8. Production quality gates and operating targets

Area	Minimum gate before commercial production	Example target
Availability	Multi-AZ services, health-based replacement, tested rollback and failover	Start with 99.9% monthly for core API; increase only with evidence and support capacity.
Recovery	Encrypted PITR plus quarterly full restore and audit-chain verification	RPO ≤ 15 minutes; RTO ≤ 4 hours for initial pilots, contractually reviewed.
Performance	Representative tenant/data load tests and slow-query budget	p95 read API < 500 ms and mutation < 1 s excluding deliberate approvals/external calls.
Security	Threat model, SAST/SCA/secret scanning, SBOM, signed artifacts, penetration test	Zero unresolved critical/high exploitable findings at release.
Tenancy	Automated two-tenant leak probes for every schema/policy change	Zero cross-tenant visibility at API and SQL layers.
Data correctness	Property/invariant tests for inventory, GL, tax, payroll, approvals and numbering	No ledger mutation outside the canonical transactional service.
Accessibility	Keyboard, focus, screen-reader and contrast tests for critical workflows	WCAG 2.2 AA target for the customer-facing and core operational paths.
AI	Feature-specific eval, grounding, privacy review, shadow mode, kill switch and rollback	No model-written financial/inventory transaction; 100% consequential actions attributable to approval.

Operational runbooks required

Platform

- Database saturation/failover
- Queue backlog and poison message
- Keycloak outage/signing-key rotation
- Bad migration and application rollback
- Object store or KMS denial

Business integrity

- Broken stock/GL invariant
- Duplicate statutory submission
- Audit-chain verification failure
- Tenant isolation incident
- AI kill switch and provider incident

9. What not to add yet

Tempting stack	Why not now	Adopt when…
Kubernetes/EKS	Three primary workloads do not justify cluster platform operations.	There are many independently owned workloads, GPU scheduling or multi-cluster policy needs.
Microservices	Would distribute transactions and debugging before domain/product boundaries have stabilized.	A measured module-level scaling, release or isolation constraint exists.
Kafka/MSK	BullMQ/outbox is adequate for current volume and simpler to operate.	Replayable high-volume streams, many consumer teams or analytics fan-out require it.
Separate vector database	pgvector is co-located with permissions and sufficient for the first authorized RAG workloads.	Index size/latency/recall measurements exceed PostgreSQL's practical envelope.
General-purpose agent framework	XELOR already has a domain-bounded graph runtime with stronger approval ancestry than generic wrappers provide.	A framework supplies a concrete missing capability without weakening controls or duplicating state.
Data lake/warehouse on day one	Creates governance and duplicated-data burden before operational query load is known.	BI workloads harm ERP performance or historical/cross-tenant analytics has a funded owner.

> **A "perfect running product" is not a stack list.** It is an architecture that stays understandable, restores correctly, preserves business invariants, exposes its failures, and can be changed safely. The recommendation above optimizes for those properties.

10. Evidence and reference basis

Repository evidence inspected

- `package.json` and all workspace package manifests
- `infra/docker-compose.yml`, Keycloak realm/theme and PostgreSQL initialization
- `infra/railway` Dockerfiles, service descriptors, startup scripts and five-container parity topology
- `packages/db` schemas, migrations, RLS/permission checks and demo reset
- `apps/api` domain modules, common guards, worker/outbox, workflow, AI and Agent OS
- `apps/web` module manifests, shell, Copilot rail and Playwright workflows
- `docs/02-investor-demo/02-capability-gaps.md`
- `docs/01-agent-os/04-managed-services.md`, shared RELAY operating model, API, UI and graph
- Module/API/deployment re-audit and Factory Connect verification run on 8 August 2026

Official external references used for future recommendations

- Amazon ECS architecture and Fargate: docs.aws.amazon.com/AmazonECS/latest/developerguide/application_architecture.html
- Amazon RDS for PostgreSQL and Multi-AZ: docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_PostgreSQL.html
- OpenTelemetry JavaScript/Node.js: opentelemetry.io/docs/languages/js/getting-started/nodejs/
- ISO/IEC 20000-1: iso.org/standard/70636.html; NIST SP 800-61 Rev. 3: csrc.nist.gov/pubs/sp/800/61/r3/final
- Google SRE service-level objectives: sre.google/sre-book/service-level-objectives/
- Temporal durable execution (future trigger, not immediate recommendation): docs.temporal.io/temporal
- Three.js post-processing and WebGPU renderer guidance: threejs.org/manual/en/post-processing.html and threejs.org/manual/en/webgpurenderer
- React Three Fiber performance scaling: r3f.docs.pmnd.rs/advanced/scaling-performance
- Browser glass and motion-accessibility primitives: MDN backdrop-filter and MDN reduced-motion guidance

Final recommendation

Keep the core stack. Invest next in managed operations, observability, recovery, security and the missing manufacturing workflows. Use the ports already present to introduce hosted AI, durable workflow or higher-scale event infrastructure only when a tested product need calls for them.

This report describes the working repository inspected on 8 August 2026. “Live” means implemented and exercised in the prototype; RELAY's service data remains illustrative and does not imply staffed coverage, a contractual SLA, an active hosted URL, production certification or regulatory approval.